Fangqiang Ding

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EDUCATION

Tongji University, Shanghai **September, 2017-June, 2021**

B.Eng in Mechanical Engineering ***GPA**: 4.77/5.00 or 92.68/100 (1/131)*

— **Scholarship**: National Scholarship × 2 (Year 2018, Year 2019) and one Hirschvogel Schorlarship

— **Award**: Top Ten Undergraduate Academic Star in Tongji University

— **Supervisor**: **Prof. Changhong Fu** (Director of **Vision4Robotics Group**)

Tsinghua University, Beijing **August, 2020-September, 2020**

Visiting Student in Department of Automation

— **Supervisor**: Prof. Geng Lu (Director of UAV Lab)

RESEARCH INTEREST

Present: Unmanned Aerial Vehicle, Computer Vision, Object Tracking

Future: 3D Mapping, Drone Odometry, Mobile Robots

RESEARCH EXPERIENCE

Improving UAV Tracking Accuracy by Regularization Modification **April, 2019-Present**

- Exploited the automatic adaptation of spatio-temporal regularization for correlation filters optimization. Local and global properties of detection-oriented response map are employed to elevate the robustness of trackers. (published by *CVPR 2020*)
- Proposed multi-view selection mechanism to leverage memory augmentation for correlation filters to alleviate exponential decay of information. Perception hash algorithm is adopted to calculate the similarity of patches and decide whether to update the memory. (published by *IROS 2020*)
- Presented target-aware dual filter learning for anti-dark UAV tracking. High-speed low-light enhancement in conjunction with pixel-level mask enable model focus on target appearance while dual filter mutually constraint to avoid filter degradation. (accepted by *ICRA 2021*)

Developing Real-Time and Robust Aerial Tracker on a Single CPU **November, 2019-Present**

- Presented distractor repressed dynamic regression tracker. The local maximum points in response map are used to localize the distractors and the regression label of objective function is dynamically

adjusted by element-wise penalization of distractors. (published by *IROS 2020*)

- Proposed joint scale and aspect ratio optimization by training separate filer in size domain. Inspired by two-stage object detection methods, a novel object proposal based re-detection strategy is presented to cope with long-term tracking tasks. (published by *IROS 2020*)
- Investigated appearance variation information to adaptively generate a spatial reliability map. A response pool is proposed as a standard of comparison to measure the stability of response map and influence its spatial reliability. (accepted by *IEEE Trans. Multimedia*)

Exploring Deep Learning Methods for Visual Object Tracking **May, 2019-Present**

- Investigated the integration of two complementary types of methods, *i.e.,* end-to-end trackers and correlation filter-based approaches to obtain CPU-friendly and high-performance trackers.
- Proposed to train domain translation model based on Generative Adversarial Networks (GANs) to generate multi-modal (RGB-T or RGB-D) tracking datasets using original visible spectrum data.
- Applied low-light enhancer in the form of multi-branch header to deep tracking paradigm to improve its adaptivity in darkness. A novel benchmark used for evaluating trackers in dark scenes is made.

Vision-Based UAV Localization under Air-Ground Robots Coordination **October, 2020-Present**

- Deigned a benchmark system that enables the evaluation of visual object tracking methods on UAV localization under air-ground robot coordination scenarios. Specifically, ground-truth positions were obtained from the motion capture system and the estimated results were output from ROS.
- Validated the effectiveness of various object trackers in accurate localization of drone. Visual trackers are assigned to predict the positions of markers in image domain which are used by classical framework of UAV pose estimation to estimate the trajectory of UAV in the world coordination frame.

PUBLICATION

Conference

- Yiming Li, Changhong Fu, **Fangqiang Ding**, Ziyuan Huang and Geng Lu. *AutoTrack: Towards High-Performance Visual Tracking for UAV with Automatic Spatio-Temporal Regularization*, published by 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR'20**). [link]
- **Fangqiang Ding**, Changhong Fu, Yiming Li, Jin Jin and Chen Feng. *Automatic Failure Recovery and Re-Initialization for Online UAV Tracking with Joint Scale and Aspect Ratio Optimization*, published by 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS'20**). [link]
- Changhong Fu, **Fangqiang Ding**, Yiming Li, Jin Jin and Chen Feng. *DR^{2}Track: Towards Real-Time Visual Tracking for UAV via Distractor Repressed Dynamic Regression*, published by 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS'20**). [link]
- Yiming Li, Changhong Fu, **Fangqiang Ding**, Ziyuan Huang and Jia Pan. *Augmented Memory for Correlation Filters in Real-Time UAV Tracking*, published by 2020 IEEE/RSJ International Conference on

Intelligent Robots and Systems (**IROS'20**). [link]

- Bowen Li, Changhong Fu, **Fangqiang Ding**, Junjie Ye and Fuling Lin. *ADTrack: Target-Aware Dual Filter Learning for Real-Time Anti-Dark UAV Tracking*, accepted by 2021 IEEE International Conference on Robotics and Automation (**ICRA'21**).
- Guangze Zheng, Changhong Fu, Junjie Ye, Fuling Lin and **Fangqiang Ding**. *Mutation Sensitive Correlation Filter for Real-Time UAV Tracking with Adaptive Hybrid Label*, accepted by 2021 IEEE International Conference on Robotics and Automation (**ICRA'21**).

Journal

- Changhong Fu, **Fangqiang Ding**, Yiming Li, Jin Jin and Chen Feng. *Learning Dynamic Regression with Automatic Distractor Repression for Real-Time UAV Tracking*, published by Engineering Applications of Artificial Intelligence. [link]
- Changhong Fu, Jin Jin, **Fangqiang Ding**, Yiming Li and Geng Lu. *Spatial Reliability Enhanced Correlation Filter: An Efficient Approach for Real-Time UAV Tracking*, accepted by IEEE Transactions on Multimedia.
- Junjie Ye, Changhong Fu, Fuling Lin, **Fangqiang Ding**, Shan An and Geng Lu. *Multi-Regularized Correlation Filter for UAV Tracking and Self-Localization*, submitted to IEEE Transactions on Industrial Electronics.
- Bowen Li, Changhong Fu, **Fangqiang Ding**, Junjie Ye and Fuling Lin. *All-Day Object Tracking for Unmanned Aerial Vehicle*, submitted to IEEE Transactions on Image Processing. [link]

Magazine

- Changhong Fu, Bowen Li, **Fangqiang Ding**, Fulin Li and Geng Lu. *Correlation Filter for UAV-Based Aerial Tracking: A Review and Experimental Evaluation*, submitted to IEEE Geoscience and Remote Sensing Magazine. [link]

SELECTED PROJECT

Zeal Eco-Power Racing Vehicle Team **October, 2018-Present**

- Designed on-board vehicle velocity measurement and automatic oil cut-off system.
- Developed mobile APP for the visualization of vehicle state parameters and driving control.
- Equipped racing vehicles with perception ability using computer vision algorithms.
- Presided over a series of training sessions for new members in electrical control group

Developing Intelligent Following Umbrella **January, 2019-May, 2019**

- Proposed to calculate pedestrian position by UWB multi-module communication.
- Exploited controlling motors directly to simulate the remote controller.
- Incorporated our designed novel umbrella for implementation.

ACADEMIC SERVICE

IROS 2020	Reviewed three submissions
IEEE JSTSRS	Reviewed one submission
IEEE TCYB	Reviewed two submission
IEEE TPAMI	Reviewed one submission
CAA PRE-IROS 2020 ONLINE	Present two papers

AWARD

- First Prize of Tongji Undergraduate Mathematics Competition (top 10) **June, 2018**
- First Prize of Shanghai College Student Innovative Design Competition (top6) **May, 2018**
- Second Prize of Shanghai College Student Physics Competition **December, 2018**
- Second Prize of Tongji Undergraduate Physics Competition **May, 2018**
- Third Prize of Chinese Undergraduate Mathematics Competition **October, 2018**
- Third Prize of Chinese Undergraduate Mathematics Modelling Competition **November, 2018**
- Third Prize of Shanghai Undergraduate Mathematics Competition **November, 2018**
- Excellent Student Award in Tongji University **December, 2018**
- Third Prize in Shanghai Mechanics Competition **May, 2019**
- Excellent Student Award in Tongji University **December, 2019**
- Top Ten Undergraduate Academic Star in Tongji University **December, 2020**

KEY SKILL

Programming Language	Matlab, Python, C++, Java, HTML
Developing&Simulation	Keil, Linux, ROS, Pytorch, Android Studio, Multisim
Tools	LaTeX, Adobe Premiere Pro, AutoCAD, Solidworks, Autodesk Inventor